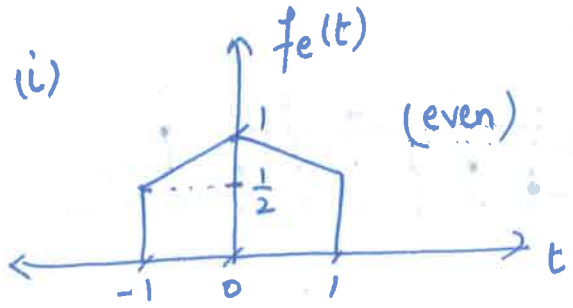
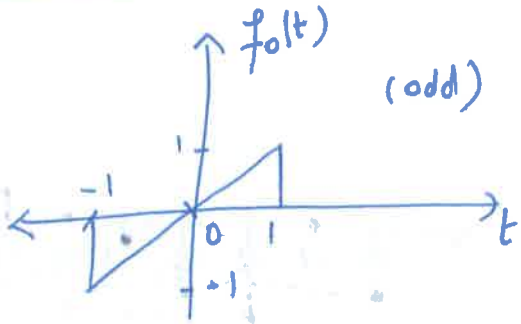
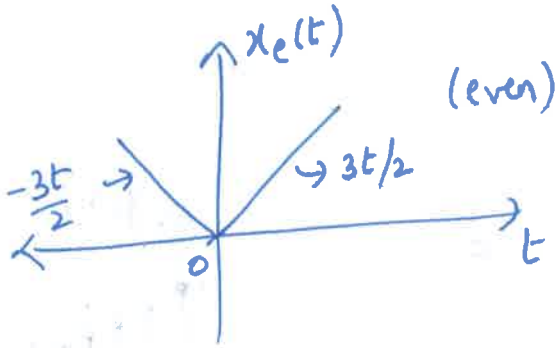
EED-201 TUTORIAL 2 : Answers

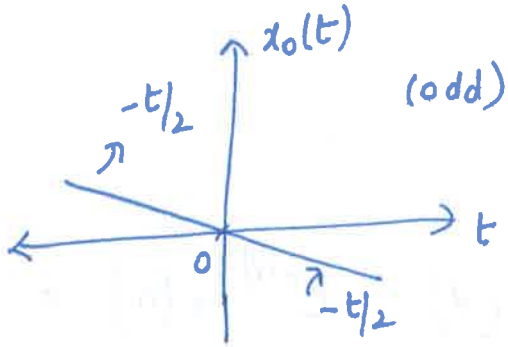
Ans. 1. (i)



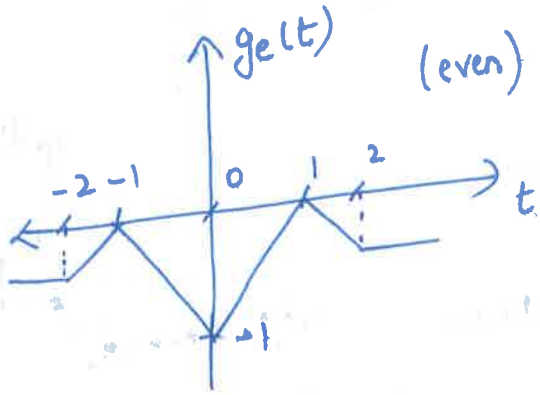


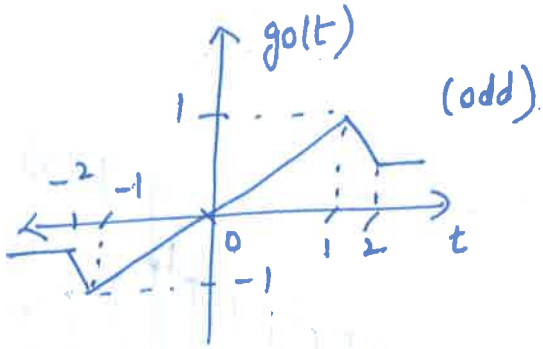
(ii)



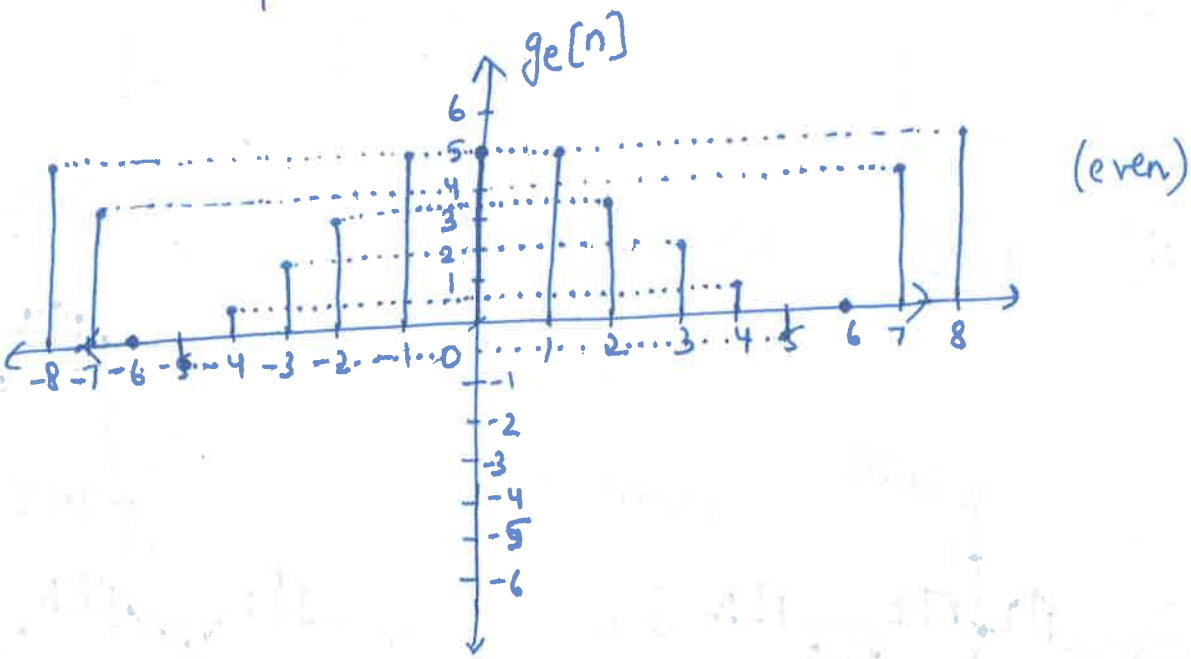


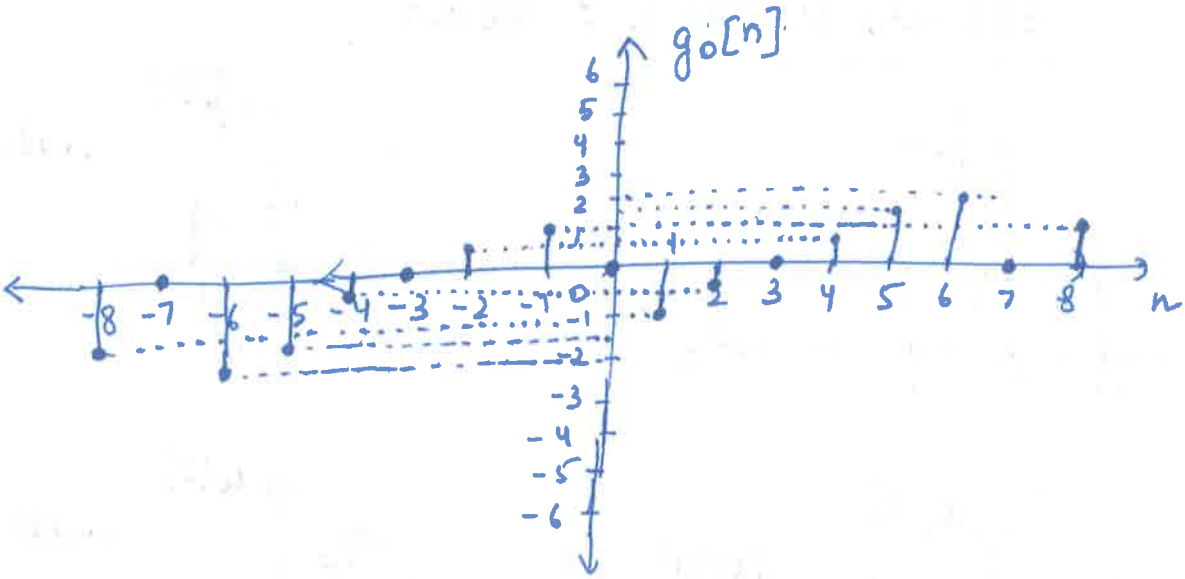
(iii)



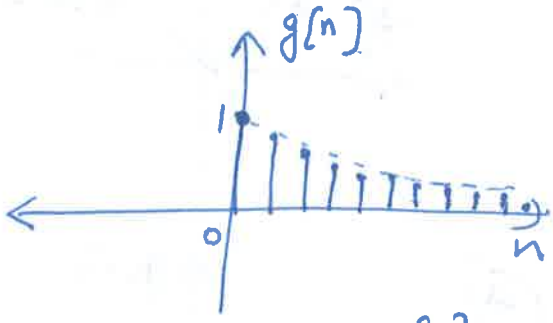


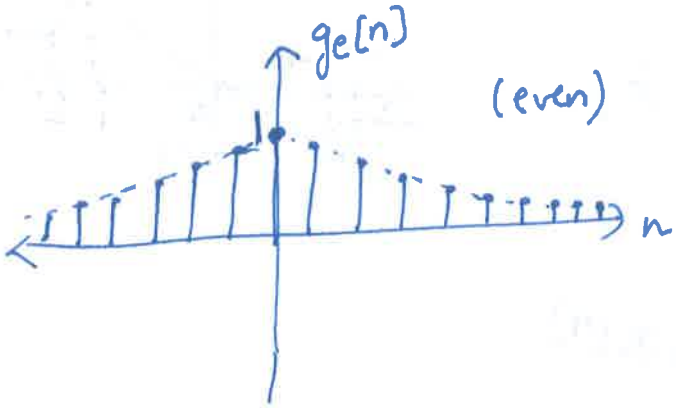
(iv)

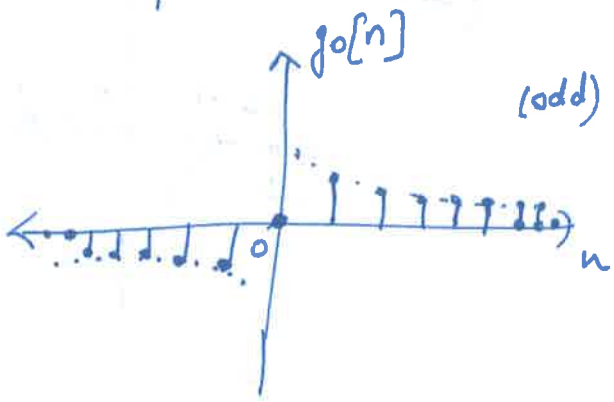




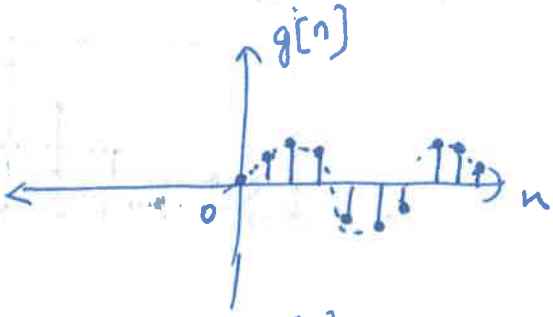
(v) $e^{-n/4}u[n] = g[n] =$

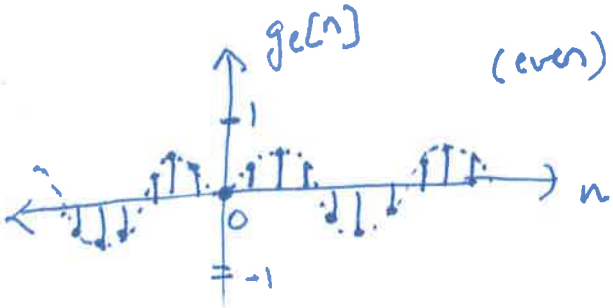


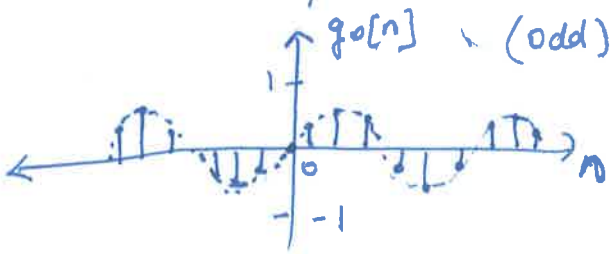




(vi) $g[n] = \sin(2\pi n/4)\,u[n]$







Ans. 2.

(i) Periodic with period = 40

(ii) Periodic with period = 252.

(iii) Non-periodic ($\because$ signal doesn't exist from $-\infty$ to ∞)

(iv) Non-periodic ($\omega/2\pi$ irrational)

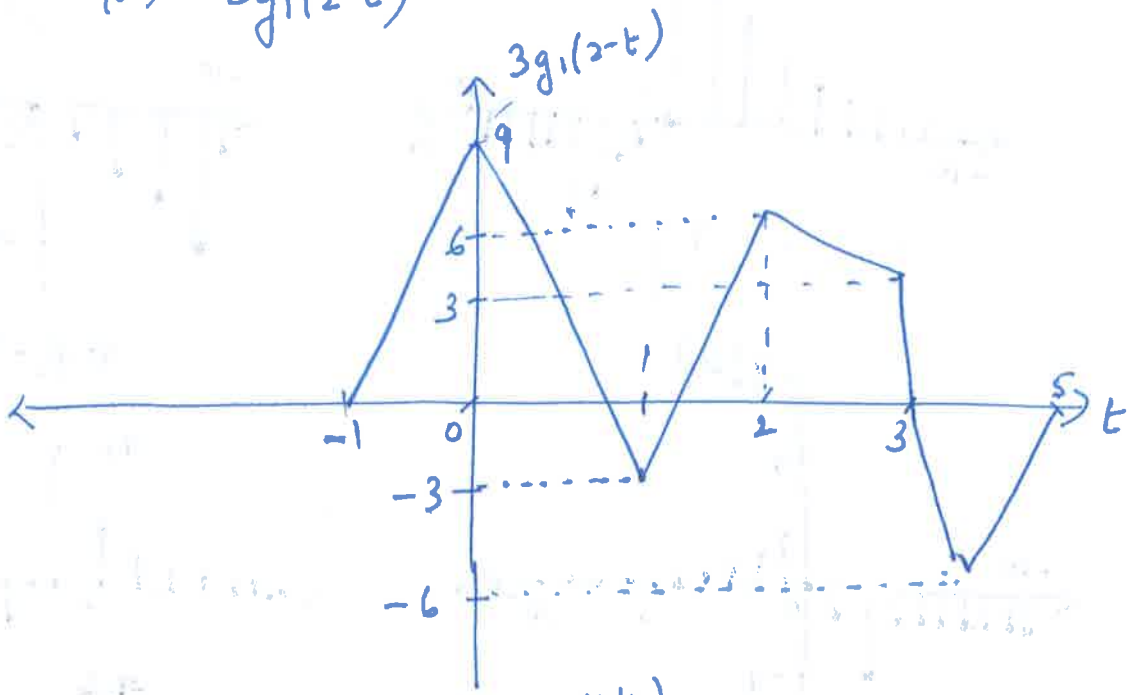
(v) Periodic with period = 8.

(vi) Periodic with period = $\pi/2$ sec.

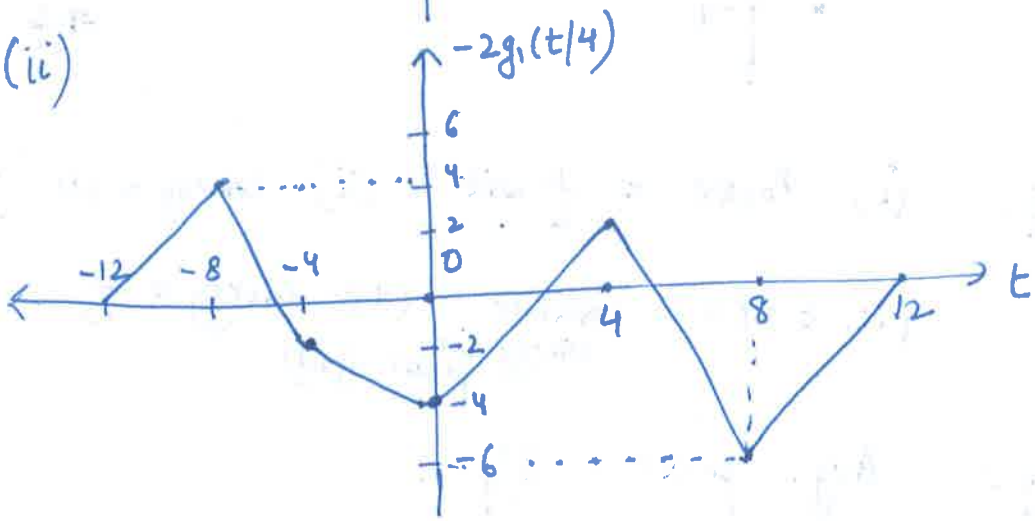
(vii) Periodic with period = π sec.

(viii) Periodic with period = 2 sec.

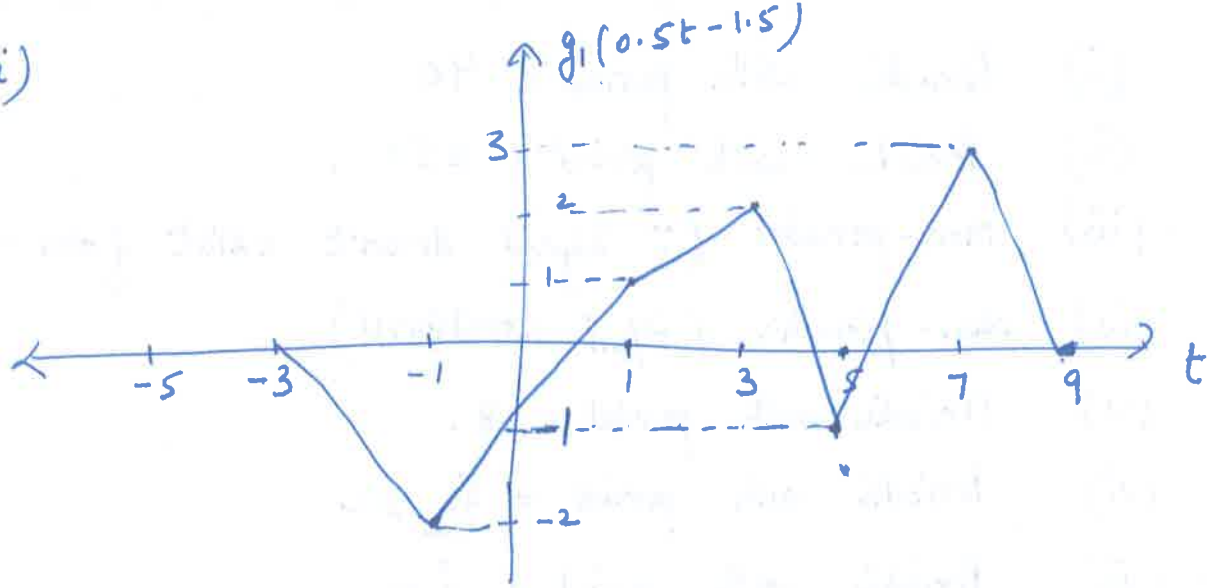
Ans. 3. (i) $3g_1(2-t)$



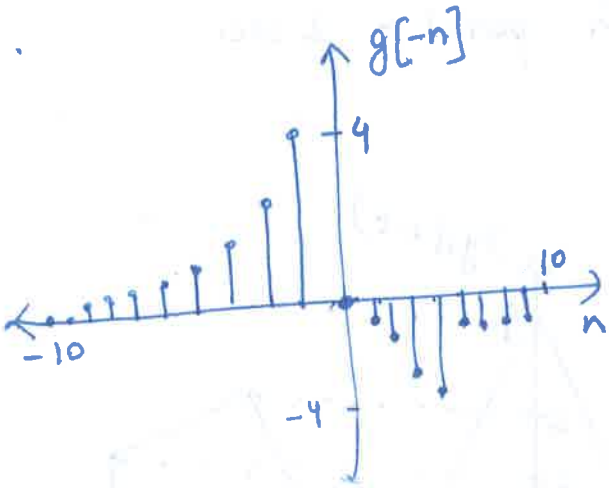
(ii)

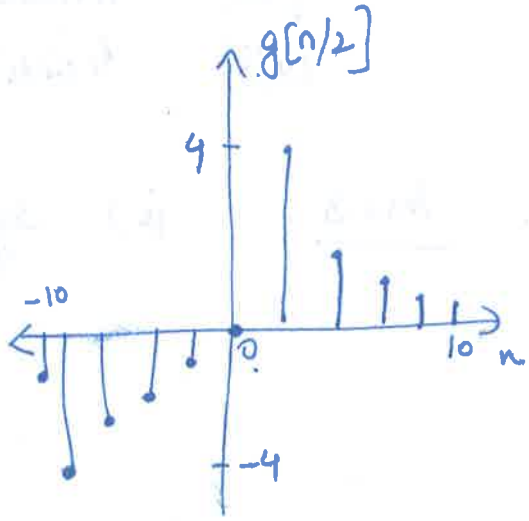


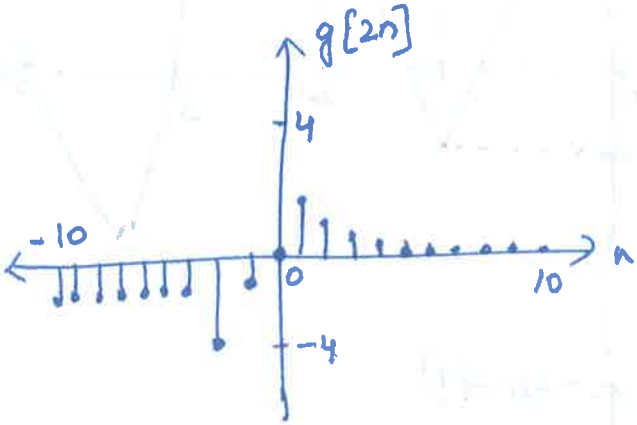
(iii)

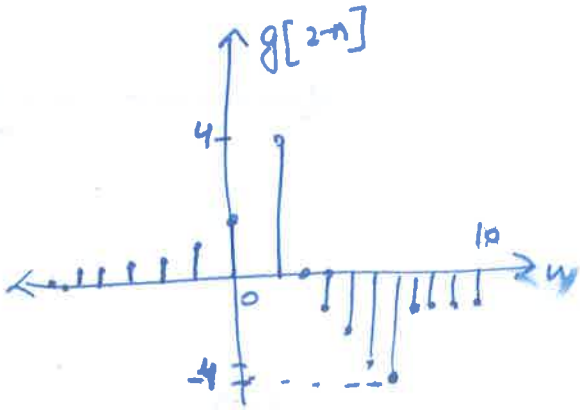


Ans-4.









Ans-5. (i) Power $= \frac{1}{2}$ units (ii) Energy $= 10$; (iii) Power $= 16$

(iv) $E = P = \infty$ (neither energy nor power signal) (v) Energy $= 2/3$.

Ans-6. Avg. power $= 8/9$.